

AMIRU MALLAWARACHCHI

AI ENGINEER | LLM APPLICATIONS • AGENTIC SYSTEMS • RAG • MODEL TRAINING

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[LinkedIn](#) • [GitHub](#) • [Hugging Face](#) • [Portfolio](#)

SUMMARY

AI Engineer who ships end-to-end: trains and deploys transformer models, then wires them into production LLM applications. Fine-tuned and published five transformer models to Hugging Face (best: macro F1 0.815, AUC 0.938) and built the async FastAPI platform that serves them — a 14-agent orchestration pipeline with a cited RAG layer, a two-stage safety gate, MongoDB persistence and SSE streaming to a Next.js frontend, backed by 730 pytest tests. Also built an MCP-based multi-agent planner over live third-party APIs, and led a five-person team whose system was chosen from Batch 124 for industry presentation at ICBT. Strong Python, comfortable owning an ambiguous problem from spec to production, and fast with coding agents (Claude Code, Cursor, Copilot).

TECHNICAL SKILLS

LLM Applications: LLM API integration (Anthropic, OpenAI, Groq), prompt & context engineering, tool / function calling, structured JSON outputs, robust response handling & retries, model routing for latency vs. reasoning depth, guardrails, prompt-injection & PII protection, LLM evaluation

Agentic Systems: Multi-step stateful agent workflows, supervisor / worker patterns, shared-state orchestration, custom async pipelines, LangGraph, LangChain, LlamaIndex, Model Context Protocol (MCP) tool servers

ML & Model Training: PyTorch, Hugging Face Transformers, transformer fine-tuning (LoRA / QLoRA / PEFT), text classification, cross-encoder rerankers, scikit-learn, XGBoost, dataset curation & class-imbalance handling, macro/micro F1, ROC-AUC, NDCG@k, MRR, threshold calibration, Hugging Face Hub publishing

RAG & Data: ChromaDB, Pinecone, FAISS, semantic chunking, hybrid search, MMR retrieval, cross-encoder reranking, citation grounding; MongoDB (schema design, indexing, query optimisation), MySQL, SQLite, Pandas, NumPy

Backend & Engineering: Python (async / asyncio), FastAPI, Pydantic, REST API design, SSE streaming, background schedulers, pytest (unit + integration), Java, TypeScript, Next.js 15, React, Tailwind CSS, SQL

Deployment & Tooling: Docker, Git / GitHub, GitHub Actions, Langfuse observability, Vercel, Cloudflare Tunnel, MongoDB Atlas; AI-assisted development with Claude Code, Cursor and GitHub Copilot

PROJECTS

MindLens — Multi-Agent AI Wellbeing Platform | Sole architect & engineer | *Python • FastAPI • PyTorch • Transformers • ChromaDB • MongoDB • Groq • Next.js • Docker*

Code: github.com/AmiruMallawarachchi/mindlens Live: mindlens-theta.vercel.app Models: huggingface.co/AmiruMallawarachchi

- **Agent architecture:** designed and built a 14-agent async orchestration pipeline in Python (asyncio, ~776-line orchestrator, no agent framework) over a shared state object — eight layers and a seven-step turn: safety gating, signal extraction, retrieval, planning, generation, post-checks and persistence, with agents fanned out concurrently and merged before generation.
- **Model training:** fine-tuned and published five transformer models to Hugging Face — a DistilBERT crisis classifier, a RoBERTa emotion classifier, a MentalBERT multi-label classifier, a RoBERTa cognitive-distortion classifier and a cross-encoder RAG reranker. Curated eight public datasets from 539,437 raw to 377,423 cleaned examples, handling imbalance up to 172x. Best model: macro F1 0.815, micro F1 0.819, average AUC 0.938.
- **Safety & evaluation:** built a two-stage crisis gate (~45 deterministic regex patterns, then the DistilBERT classifier at a calibrated 0.45 threshold). Iterated across three model versions against a held-out probe suite until v3 caught 5/5 crisis probes at every threshold from 0.45 to 0.9 with 0/6 false positives and a +0.998 separation margin — up from +0.06 in v1.
- **RAG system:** built a grounded retrieval layer over ChromaDB — a 60-entry cited corpus chunked to ~67 passages, MMR retrieval (fetch_k 20, lambda 0.5), recency boosting, then cross-encoder reranking to k=5. Trained the reranker on 5,069 mined pairs with 400 LLM-judged examples (36.8% positive) and evaluated with NDCG@3, MRR and P@3.
- **Platform engineering:** async FastAPI service with Pydantic contracts, dual-tier Groq model routing to trade latency against reasoning depth, two-layer memory (working session + durable user memory), APScheduler check-ins, MongoDB Atlas persistence, external tool integration via a public music search API, and SSE streaming to a Next.js 16 / React 19 frontend. ~11,900 lines of Python and ~18,000 of TypeScript across 180 commits, with 730 passing pytest tests, pre-commit hooks and CI, built to Sri Lanka PDPA 2022 data-protection requirements.

TripWeaver — MCP-Based Multi-Agent Travel Planner | Capstone, STEMLink AI Engineer Bootcamp 2.0 | *FastAPI • LangGraph • FastMCP • Amadeus API • Gradio*

Code: github.com/AmiruMallawarachchi/multi-agent-travel-planner Live: multi-agent-travel-planner-jet.vercel.app

- Built a supervisor/worker agent graph in LangGraph where specialist flight, hotel and itinerary agents coordinate through shared state to turn one free-text trip request into a costed, day-by-day plan.
- Exposed live Amadeus flight and hotel inventory to the agents as tools through two FastMCP servers — real Model Context Protocol tool access rather than hard-coded API calls — so tools can be swapped or added without touching agent logic.
- Shipped as a streaming Gradio product over a FastAPI service layer, with token streaming and graceful degradation when an upstream provider returns no inventory.
- Built through the STEMLink AI Engineer Bootcamp 2.0, which took me from LLM API basics to production agent design — prompt and context engineering, tool/function calling, structured outputs, LangChain and LangGraph orchestration, RAG over vector stores, MCP tool servers, and evaluation. Assessed by viva and certified as a Gen AI Engineer; this project is where most of my agentic design instincts came from.

Automated Presidential Election System | Team Leader, 5-member team — EXITO'26, ICBT | PHP 8 · MySQL 8 · Bootstrap · Docker

Code: github.com/AmiruMallawarachchi/election-system Live: election-system-pearl.vercel.app

- Led a five-member team to build an end-to-end Sri Lankan presidential election platform: voter registration, Grama Niladhari officer verification, admin-controlled election scheduling, ranked-preference voting, live turnout monitoring and result publication, across three role-based dashboards in English, Sinhala and Tamil. Split scope across five developers, reviewed merges and led the industry presentation.
- Selected from the whole of Batch 124 for the EXITO'26 Exhibition at ICBT Campus and presented to lecturers, senior students and IT industry professionals, whose feedback fed directly into the next iteration of the system.
- Owned architecture and delivery: MVC structure, MySQL schema and migrations, and a security layer with CSRF tokens, prepared statements, password hashing, OTP + PIN authentication, session regeneration, failed-login tracking and one-vote-per-voter enforcement — the same threat-modelling habit I now apply to prompt injection and PII exposure in LLM systems.

ADDITIONAL ML WORK

- **Kaggle Playground Series S6E6 — Stellar Classification:** multi-class GALAXY / STAR / QSO classification; feature engineering and XGBoost hyperparameter tuning to a 0.95458 leaderboard score.
- **Multi-Task NLP Service:** a five-pipeline Hugging Face inference API (summarisation, sentiment, zero-shot classification, NER, extractive QA) behind a typed FastAPI layer with Pydantic schemas.
- **DocuMind AI:** LangGraph ingestion and retrieval pipeline over a Pinecone vector store with grounded, cited question answering, streamed to a Next.js 15 frontend over SSE.
- **Technical writing:** explainers on transformer architecture and sequential modelling, published on Medium and LinkedIn under "Amiru AI Lab".

EXPERIENCE

Web Development Intern — Syntexhub (Remote) Jan 2026 – Feb 2026

- Built and shipped React / MERN features against an existing codebase, working remotely to ticket-level specs and code review.

Customer Service Executive — Bellvantage (Pvt) Ltd, Colombo Jul 2023 – Feb 2024

- Supported Sampath Bank's financial recovery operations, handling 60–80+ inbound and outbound calls daily under banking compliance and data-privacy rules.
- Logged interactions and payment agreements in internal CRM systems and analysed account patterns to support recovery strategy — where I first learned to handle sensitive personal data carefully.

EDUCATION

BSc (Hons) Software Engineering — Cardiff Metropolitan University (ICBT Campus, Colombo) Expected Dec 2026

Higher National Diploma in Computing (Merit) — ICBT Campus, Colombo Completed

GCE Advanced Level — St. Sebastian's College, Moratuwa 2019

CERTIFICATIONS & ACHIEVEMENTS

- Gen AI Engineer — STEMLink AI Engineer Bootcamp 2.0 (Aug 2026)
- Introduction to LangChain — LangChain Academy (Apr 2026) · Python (Basic) — HackerRank (Apr 2026) · Intro to Machine Learning — Kaggle (2026)
- Team Leader, EXITO'26 Exhibition — project selected from Batch 124 for presentation to industry professionals, ICBT Campus (Jul 2026)
- **Languages:** English (bilingual proficiency) · Sinhala (native)